

Detailed Milestone 4 Report

Multiple-Choice Classification using BERT and LoRA

Introduction

This milestone formulates the Smart MCQ Solver Challenge as a supervised multiple-choice classification task using Transformer models. Each question contains one prompt and five candidate answers. The goal is to predict the correct option by learning contextual relationships between the prompt and every answer choice.

Objectives

Convert labels to numeric form; format prompt-option pairs; tokenize data using bert-base-uncased; build tensors for multiple-choice learning; use AutoModelForMultipleChoice; apply LoRA for efficient fine-tuning; prepare datasets for Hugging Face Trainer; perform a small fine-tuning experiment; compute probabilities using Softmax.

Dataset

Each row contains a prompt, options A-E, and an answer column. Labels are encoded as A=0, B=1, C=2, D=3, E=4. Each question becomes five prompt-option sequences.

Preprocessing

Every option is concatenated with the prompt using the separator token: Prompt [SEP] Option. Tokenization uses padding='max_length', truncation=True and a fixed maximum length. Output tensors include input_ids and attention_mask.

Model Architecture

AutoModelForMultipleChoice uses a pretrained BERT encoder. Each option is encoded independently, pooled, and passed through a classification layer. The model outputs five logits, one for each answer option.

Training Loss

Cross-entropy loss compares the predicted logits with the encoded ground-truth label. During optimization only the correct answer receives the positive target while the remaining options are treated as negatives.

LoRA

Low-Rank Adaptation freezes most BERT parameters and inserts trainable low-rank matrices into attention query and value layers. Configuration: r=8, alpha=16, dropout=0.1, bias='none', task_type='SEQ_CLS'. This dramatically reduces trainable parameters, memory usage and training time.

Dataset for Trainer

The processed dataset stores `input_ids` with shape `[5,128]`, `attention_mask` with shape `[5,128]`, and one numeric label for every question. This format is directly accepted by Hugging Face Trainer.

Fine-Tuning

A small experiment trains the LoRA model using a batch size of four, maximum sequence length of sixty-four tokens and four optimization steps. The purpose is to validate the end-to-end pipeline rather than maximize accuracy.

Inference

After training, logits are produced for the five answer choices. Softmax converts logits into probabilities whose sum equals one. The option with the highest probability becomes the prediction.

Advantages

Multiple-choice models evaluate all answer options jointly, allowing the model to compare semantic differences directly. LoRA enables efficient adaptation with far fewer trainable parameters than full fine-tuning, making experimentation faster and cheaper.

Conclusion

This milestone demonstrates a complete workflow for building a Transformer-based MCQ solver, including preprocessing, tokenization, multiple-choice modeling, LoRA fine-tuning, Trainer integration and probabilistic inference. The techniques form a practical foundation for efficient large language model adaptation on educational question-answering tasks.

References

Hugging Face Transformers Documentation; Hugging Face Multiple Choice Task Guide; PEFT/LoRA Documentation; PyTorch Documentation; Smart MCQ Solver Challenge (Kaggle).